
Project Management Plan

for

Escape Sequence

Version 1.0 draft 1

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1. Overview

Escape Sequence is a Java-based software engineering project developed by Akera Griffith and Kaitlyn Morris for CS 491 and CS 492 under Dr. Zhao. The project is a 2D space-themed survival card game inspired by Blackjack, where players must survive three increasingly difficult rounds in order to reach limited escape pods. The game combines traditional card mechanics with specialty cards, progressive difficulty, and a thematic graphical user interface.

The motivation for this project is to create an engaging and interactive game that applies software engineering concepts in a practical way. The project allows the team to strengthen skills in object-oriented programming, GUI design, game logic, documentation, testing, and project planning. The final product is intended to demonstrate the team's ability to design, implement, test, and document a complete software system.

Major milestones for the project include requirements gathering, design documentation, interface design, core game logic implementation, testing, debugging, and final presentation. The project is constrained by the academic semester timeline, available team resources, and the technical limitations of implementing all desired features within the course schedule.

1.1. Project Purpose, Objectives, and Success Criteria

The purpose of this project is to design and develop a playable software product called Escape Sequence, which combines Blackjack-style gameplay with a survival-based spaceship theme. The project supports both single-player and multiplayer modes, records player outcomes, and provides an interactive graphical interface with instructions and specialty card mechanics.

The main objectives of the project are:

- To design and implement a functional Java-based 2D card game.
- To create a user-friendly GUI that supports core gameplay actions such as hit, stay, and specialty card use.
- To develop clear game rules, player progression, and round-based logic.
- To support up to six players in multiplayer mode.
- To maintain local storage of player names, results, and scores.

The success criteria for the project are as follows:

- The game launches and runs successfully on standard computers with Java installed.
- Players can complete a full game from start to finish without major errors or crashes.
- Core features such as title screen, player name entry, round progression, hit, stay, specialty cards, and results recording function correctly.
- The system responds to user inputs within acceptable time limits.
- The software is documented with the required proposal, SRS, diagrams, and project management materials.
- The project is completed and submitted within the deadlines established for the course.

This project is related to the course objectives of CS 491 and CS 492, and integrates concepts from prior programming and software design experience. Shared resources for the project include the development environment, Java libraries, design notes, and project documentation. Feature dependencies include the need to complete game logic before fully integrating GUI components and testing.

1.2. Project Deliverables

Deliverable	Recipients	Delivery Date	Delivery Method	Comments
Project Proposal	Dr. Zhao	September 9, 2025	Course submission	Initial concept and project scope
Software Requirements Specification (SRS)	Dr. Zhao	October 20, 2025	Course submission	Includes requirements, diagrams, and interfaces
Use Case Diagram	Dr. Zhao	With SRS	Embedded in document	Shows player-system interactions
Sequence/Activity Diagrams	Dr. Zhao	With SRS	Embedded in document	Supports analysis and design
GUI Mockups / Interface Screens	Dr. Zhao	During design phase	Embedded in documentation	Illustrates look and feel of game

Working Prototype	Dr. Zhao	Mid-to-late semester	Demo / course submission	Demonstrate partial implementation
Final Game Application	Dr. Zhao	End of semester	Digital submission	Completed playable version
Testing and Debugging Results	Dr. Zhao	End of semester	Attached report	Summary of testing outcomes
Final Project Presentation / Demo	Dr. Zhao / Class	End of semester	In-class presentation	Demonstration of features and results
Final Documentation Package	Dr. Zhao	End of semester	Course submission	Includes revised documents and supporting materials

1.3. Assumptions, Dependencies, and Constraints

Assumptions

- **AS-1:** All team members will have access to laptops or computers capable of running Java development tools.
- **AS-2:** The project will be developed using Java and NetBeans as planned.
- **AS-3:** Players will have basic knowledge of using a keyboard and mouse.
- **AS-4:** The final software will run in an environment where Java Runtime Environment is installed.
- **AS-5:** Team members will be able to communicate regularly and divide work fairly.

Dependencies

- **DE-1:** The project depends on the availability of Java JDK, JRE, and NetBeans IDE.

- **DE-2:** GUI development depends on Java Swing libraries functioning properly.
- **DE-3:** Testing depends on completion of the core game logic.
- **DE-4:** Multiplayer implementation depends on the team deciding on and completing a workable design.
- **DE-5:** Documentation depends on the completion of requirements, design, and implementation details.

Constraints

- **CO-1:** The project must be completed within the Spring 2026 course timeline.
- **CO-2:** The project is limited to the technical knowledge and experience of a two-person student team.
- **CO-3:** The software must be developed using available academic resources and without requiring costly tools.
- **CO-4:** Time limitations may restrict the level of polish or the complexity of optional features.
- **CO-5:** The project must satisfy course requirements for documentation, design, and implementation.

Of the five project dimensions:

- **Schedule** is a major constraint.
- **Features** are a major success driver.
- **Quality** is a success driver.
- **Staff** is fixed at two team members and is therefore constrained.
- **Cost** has flexibility because the project uses academic tools and free resources.

1.4. References

IEEE Std 830-1998, *IEEE Recommended Practice for Software Requirements Specifications*.

CS 491 Course Materials, Dr. Zhao, Western Illinois University, Fall 2025.

Escape Sequence Project Proposal, The Space Cadets, September 9, 2025.

Escape Sequence Software Requirements Specification, Version 2.0, October 20, 2025.

Oracle Java SE Documentation.

NetBeans IDE Documentation.

Resident Evil 7 DLC “21” and *Mouthwashing* as thematic inspiration sources.

1.5. Definitions and Acronyms

SRS – Software Requirements Specification. A formal document describing the functional and nonfunctional requirements of the software system.

GUI – Graphical User Interface. The visual interface through which users interact with the software.

JDK – Java Development Kit. A software development environment used to build Java applications.

JRE – Java Runtime Environment. The environment required to run Java applications.

JVM – Java Virtual Machine. The platform that executes Java bytecode.

IDE – Integrated Development Environment. A software application that provides programming tools in one environment.

TCP/IP – Transmission Control Protocol / Internet Protocol. A communication protocol that may be relevant if multiplayer networking is implemented.

Host Player – The player in multiplayer mode who starts the session and controls shared setup options.

Specialty Card – A special in-game card that gives an effect different from normal numbered cards.

Keycard – A game item earned by winning a round, allowing progress toward the escape pods.

1.6. Evolution of the Plan

This project management plan will be reviewed and updated throughout the life of the project as new information becomes available. Scheduled updates will take place after major milestones such as proposal approval, SRS completion, design review, implementation progress checks, and testing. Unscheduled updates may occur if project requirements, team responsibilities, technical decisions, or deadlines change.

The latest version of the plan will be maintained as part of the team's project documentation. Each revision will include the revision date, summary of changes, and updated version number. The baseline version will be the first approved plan submitted for the course. Future versions will be stored in the project files and shared between team members to ensure both members are working from the most current version.

Configuration control for this document will be managed by keeping dated versions of the file and updating the revision history when major changes are made. This will help the team track project decisions, changes in scope, and progress across the semester.

2. Project Organization

2.1. External Interfaces

The Escape Sequence project interacts with several external interfaces throughout the development process. These external interfaces help guide the project requirements, ensure quality, and support project evaluation.

Course Instructor (Project Supervisor – Dr. C. Zhao)

The primary external interface for the project is the course instructor, Dr. Zhao. The instructor acts as the project supervisor and stakeholder responsible for reviewing deliverables, providing feedback, and ensuring that the project meets the course requirements. Communication with the instructor occurs through scheduled class meetings, assignment submissions, and project presentations.

Course Administration and Academic Environment

The project operates within the academic environment of Western Illinois University. Project documentation and deliverables are submitted through Western Online or other required submission platforms.

End Users (Players)

Although the project is primarily an academic exercise, the intended users of the software are players who interact with the Escape Sequence game. These users represent individuals with little or no technical

background who want to play an entertaining card-based survival game. Their interaction with the system influences usability requirements, and interface design.

Software Tools and Development Platforms

The project also depends on external software tools such as the Java Development Kit (JDK) and NetBeans IDE, which provide the development environment required to build the application. These tools support compilation, debugging, testing, and code organization.

Because this is a small academic project, the system does not involve external organizations such as marketing, finance, manufacturing, or procurement departments. No subcontractors are used in the development process.

2.2. Internal Structure

The internal structure of the Escape Sequence project consists of a small two-person development team working collaboratively on all aspects of the software engineering process. Responsibilities are shared between both team members, but each member contributes to specific areas based on strengths and task assignments.

The development team is responsible for designing, implementing, testing, and documenting the Escape Sequence game. Because the team is small, communication between members occurs frequently and directly to coordinate tasks and track progress.

Communication within the Escape Sequence development team occurs through several methods to ensure effective collaboration and coordination throughout the project lifecycle. The primary methods include regular in-person meetings where team members discuss project progress, assign tasks, review development work, and plan upcoming milestones. The team also uses shared project documents to allow both members to access and update project artifacts such as requirements documents, diagrams, and planning materials. In addition, informal planning discussions during development help the team quickly resolve design questions, coding issues, and implementation decisions.

Major project decisions, including design changes, gameplay mechanics, and system architecture updates, are discussed collaboratively between team members before implementation to ensure both members agree on the approach and understand the impact of the changes on the overall system.

2.3. Roles and Responsibilities

The Escape Sequence project is developed by The Space Cadets team, consisting of Kaitlyn Morris and Akera Griffith under the supervision of Dr. Zhao. Because the project team is small, each member takes on multiple roles to ensure that all project tasks are completed efficiently.

Role	Individual	Responsibilities
Project Manager	Kaitlyn Morris	Oversees the overall project schedule, coordinates team tasks, tracks progress, and ensures that milestones and deliverables are completed on time.
Software Engineer	Kaitlyn Morris	Develops core game logic, implements gameplay mechanics, and contributes to GUI functionality and system implementation.
Subject Matter Expert	Kaitlyn Morris	Provides expertise on Blackjack mechanics and gameplay rules used to design the Escape Sequence game system.
Technical Lead	Akera Griffith	Guides technical decisions, organizes program structure, and ensures that coding practices follow proper design principles.
Systems Engineer	Akera Griffith	Ensures that all components of the system (game logic, GUI, and data storage) integrate correctly.
Test Engineer	Akera Griffith	Responsible for testing system functionality, identifying bugs, and verifying that the game operates according to the requirements.
Quality Assurance Coordinator	Shared Responsibility	Both team members review documentation, verify system behavior, and ensure that the software meets project requirements.

Stakeholders

Stakeholder	Role in Project	Importance
Dr. Zhao (Instructor)	Project supervisor who reviews deliverables, provides feedback, and evaluates project progress.	High
Akera Griffith	Developer and technical contributor responsible for implementation and testing.	High

Kaitlyn Morris	Developer and project manager responsible for coordinating tasks and implementing features.	High
End Users (Players)	Individuals who interact with the game and influence design decisions related to usability and gameplay experience.	Medium
Course Environment (CS 492)	Provides project requirements, deadlines, and evaluation criteria.	Medium

Stakeholder Responsibilities by Project Phase

Phase	Stakeholder Involvement
Planning	Team defines project concept and proposal; instructor reviews and approves project scope.
Requirements	Team develops system requirements and SRS; instructor provides feedback.
Design	Team creates system architecture and gameplay design.
Implementation	Developers implement the system and integrate software components.
Testing	Team performs testing and debugging to verify system behavior.
Final Delivery	Instructor evaluates final product and documentation.

Stakeholder Relationships

The development team collaborates closely throughout the project, sharing responsibilities for design, coding, and testing. The instructor provides oversight and evaluation while guiding the team toward meeting course objectives. End users influence system usability and gameplay design but do not participate directly in development.

Stakeholder Participation Resources

To ensure effective participation in the project, the following resources are required:

- Access to development tools such as Java JDK and NetBeans IDE.
- Course materials and guidance provided by Dr. Zhao.
- Time allocated for development meetings and collaboration.
- Documentation templates and project guidelines

Stakeholder Participation Schedule

Stakeholder	Participation Period
Dr. Zhao	Proposal review, SRS evaluation, milestone reviews, final presentation
Akera Griffith	Entire project lifecycle
Kaitlyn Morris	Entire project lifecycle
End Users	Considered during design and usability evaluation

3. Managerial Process Plans

3.1. Start-Up Plans

3.1.1 Estimation Plan

Project estimates were determined through a combination of task decomposition and team judgement. The team broke the project into major function areas: Core Game Mechanics, User Interface, Settings & Additional Features, Multiplayer & Data, Testing, and Documentation. Then estimated the duration of each task. Estimations were prepared by both team members collaboratively during project development and were scheduled using a gantt chart. The team works in bursts rather than scheduled hours, so tasks durations are estimated based on approximate days needed for a task. Subtasks that have been assigned to an individual team members as an estimated duration in hours. Contingency is built into the schedule by sequencing testing phases toward the end of the project. This allows earlier phases to run slightly past planned duration without impacting the end date.

3.1.2 Staffing Plan

The project team consist of two members:

Team Member	Role	Responsibilities
Kaitlyn Morris	Project Manager / Developer	Project oversight, scheduling, documentation organization, design, development, testing
Akera Griffith	Head Developer	Development, documentation, implementation, testing

Both team members are available for the duration of the project, August 2025 - May 2026, Neither member is from an outside source; both are internal to the project. There is no possibility of another member being added to the team. New tasks will be determined and assigned at the beginning of each development phase.

3.1.3 Staff Training Plan

Both team members are computer science students in a software engineering class and are proficient in Java. The following training needs have been identified based on possible project issues:

Training Needed	Who	Method
Java Networking/Multiplayer	Both	Self-directed research, online documentation, tutorials.
Github version control workflow	Both	GitHub documentation and practical use

Due to the academic nature of the project, training is expected to occur on an as needed basis during the development phase. There is not formal training planned at this time.

3.1.4 Resource Acquisition Plan

All resources required for this project are either freely available, so no procurement or purchasing is required.

Resource	Type	Source
NetBeans IDE	Development Tool	Apache NetBeans (Free)
Java JDK	Development Tool	Oracle/OpenJDK (Free)
GitHub	Version Control	Free
Jira	Scheduling Tool	Free
Personal Computers	Hardware	Team-owned

No shared test facilities or special hardware is required. The final product will run on a standard desktop or laptop computer with java runtime environment installed.

3.1.5 Project Commitments

Commitment	Made By	Made To	Due Date	Comments
Complete Core Game Mechanics	Both	Dr. Zhao	March 23, 2026	
Complete User Interface	Both	Dr. Zhao	April 5, 2026	
Complete Multiplayer	Both	Dr. Zhao	April 19, 2026	
Complete Testing	Both	Dr. Zhao	May 3, 2026	
Final Project Delivery	Both	Dr. Zhao	May 10, 2026	

Any changes to commitments will be communicated between team members via GitHub and discussed at the next team meeting

3.2. Work Plan

The work breakdown for Escape Sequence is organized into six major phases as described in the project Gantt chart. Tasks are listed below with estimated durations and assigned responsibilities.

Phase	Task	Duration	Assigned To	Status
Core Game Mechanics	Deck Creation	9 days	Akera	Not Started
	Game Logic	12 days	Kaitlyn	Not Started
	Specialty Card System	7 days	Kaitlyn	Not Started
	Player Classes	9 days	Akera	Not Started
	Keycard/ P.A.C Mechanics	7 days	Kaitlyn	Not Started
User Interface	Button Mechanics	14 days	Akera	Not Started
	Additional Features	23 days	Both	Not Started
Settings & Additional Features	Settings Fncionality	16 days	Akera	Not Started
	Create Tutorial	16 days	Kaitlyn	Not Started
Multiplayer & Data	Multiplayer Networking	21 days	Both	Not Started

	Local Storage	14 days	Kaitlyn	Not Started
Testing	Game Logic Testing	12 days	Both	Not Started
	Integration Testing	9 days	Both	Not Started
	Multiplayer Testing	15 days	Both	Not Started
Documentation & Delivery	Complete Project Deliverables	7 days	Both	In-Progress

Major Milestones

Milestone	Target Date
Core Game Mechanics Complete	March 23, 2026
User Interface Complete	April 5, 2026
Settings & Additional Features Complete	April 19, 2026
Multiplayer & Data Complete	April 25, 2026
All Testing Complete	May 3, 2026
Final Delivery	May 10, 2026

The game narrative and screen designs have already been completed and serve as the foundation for the user interface and mechanics development phases. Tasks may be adjusted as project progresses.

3.3. Control Plan

<This section describes how the project will control and report on the project status and activities. Specify the frequency at which the various project status indicators are to be monitored and specific events that could trigger a status evaluation.>

3.3.1 Data Control Plan

Project data consists of source code, design documents, the project management plan, software requirements specification and supporting materials. The following controls are in place:

Source code is managed via GitHub, which provides version history, change tracking and rollback capability. All code changes must be committed with descriptive commit messages.

Documentation is stored in Google Docs, accessible to both team members. The Jira project space will be continuously updated with new tasks and sprints as the project continues. The game application itself contains no sensitive or personal data. Game state data will be managed in local memory during runtime with no persistent storage required.

3.3.2 Requirements Control Plan

The core requirements for Escape Sequence are defined in the SRS document for the project. If a requirement change is identified (like any additional game mechanics) the following process will be followed: the team member identifying the change will inform the other member, the team agrees on the change of requirements, the impact on the schedule and effort estimations will be reassessed and the proper documentation updated to reflect the change. Any major changes will be documented in the revision history of the Project Management Plan and Software Requirement Specification document.

3.3.3 Schedule Control Plan

Schedule process will be tracked using Jira and the projects Gantt chart. At each milestone completion, both team members will review actual progress against the planned schedule. If a task runs more than three days behind its planned end date, the team will assess the impact and adjust downstream tasks accordingly. If the multiplayer networking phase, which is the highest risk phase, runs significantly behind schedule, the team may consider reducing the multiplayer scope by reducing the maximum players or removing the functionality entirely. Any changes will be reflected in the Gantt chart and revision history.

3.3.4 Budget Control Plan

This project has no monetary budget as it is an academic project using free tools and team-owned hardware. The primary resource being managed is time. The project manager is responsible for monitoring time spent on each phase against estimates and raising concerns if a phase is consuming significantly more effort than planned. If effort overruns are identified, the team will discuss scope reduction or tasks redistribution at the next check-in.

3.3.5 Communication, Tracking, and Reporting Plan

Type of Communication	Communication Schedule	Typical Communication Mechanism	Who Initiates	Recipient
Progress Report	every Monday	In-Person or messaging	Both	Both members
Code Updates	Ongoing	GirHub Commits	Both	Both Members
Schedule and Effort Tracking Report	At Milestone Completion	Gantt chart or Jira update	Project Manager	Other Team Member
Issue Flagging	As issues arise	Messaging/ Github	Either Member	Other Team Member
Risk Mitigation Status	as mitigation actions are completed	email	responsible team member	Project Manager
Requirement Changes	as changes are approved	email and change control tool	Both	Project Manger
Instructor Updates/ Submissions	Per course schedule	Email or WesternOnline	Project manager or both	Dr. Zhao

Given the small team size, communication is expected to be informal and frequent rather than structured through formal reports.

3.3.6 Metrics Collection Plan

The following metrics will be tracked throughout the project to monitor progress and quality:

Metric	Collection Method	Frequency	Analyzed By
Task completion percentage	Gantt chart or Jira space	At each milestone	Project Manager
Number of GitHub Commits	GitHub Repository	Weekly	Both
Number of bugs/issues found during testing	SRS document or GitHub Issues	During testing phase	Both
Number of bugs/issues resolved	GitHub Issues	During testing phase	Both
Schedule Variance	Gantt chart comparison	At each milestone	Project Manager

Metrics will be reviewed informally at each milestone and used to inform schedule and scope decisions going forward.

3.4. Risk Management Plan

The following risks have been identified for the Escape Sequence project. Each risk is assigned a likelihood and impact rating (Low/ Medium/ High) and a mitigation strategy.

ID	Risk	Likelihood	Impact	Mitigation
R-1	Multiplayer networking is too	High	High	Begin research early; scope down number of players if necessary; highest priority risk

	complex for the available time frame			
R-2	Team works sporadically which may cause tasks to fall behind	Medium	High	Set clear milestone deadlines; Project manager monitors progress and modifies schedule if needed
R-3	One team member becomes unavailable due to personal reasons	Low	High	Maintain well-commented, up-to-date GitHub repository so either member can continue the others work.
R-4	Game logic bugs	Medium	Medium	Dedicated game logic testing phase; clear rule guide before implementation
R-5	Scope becomes too large (teams gets hung up on non-necessary additional features.	Low	Medium	Project manager ensures tasks are being completed on time; additional features only to be added after main tasks are complete.

Risks will be reviewed at each milestone. If a new risk is identified at any point, it will be added to this table and a mitigation strategy assigned. R-1 multiplayer implementation is the most critical to the projects success and will be monitored closely.

3.5. Issue Resolution Plan

Issues that arise during the project will as follows. Any team member who identifies an issue, whether technical, scheduling-related or interpersonal, is responsible for raising it promptly to the other team member. Technical issues will be logged as GitHub issues and assigned to the appropriate team member for resolution. Schedule or scope issues will be discussed with the project manager to determine if the needs to be a change in priority or scope changes. If the team members disagree on a course of action, the matter will be discussed with Dr. Zhao for guidance.

All open issues will be tracked in GitHub and reviewed at each milestone check-in. An issue is considered resolved when the fix has been implemented, tested, and committed to the repository, or when a formal decision has been made to defer or descope the related feature.

3.6. Project Close-Out Plan

Upon completion of the project in May 2026, the following close-out activities will be performed: All source code will be finalized, and committed to the GitHub repository with a clearly tagged release version. All project documentation, including this Project Management Plan, will be reviewed for its completeness and submitted per course requirements. The team will complete the necessary project documentation as dictated by Dr. Zhao. The project code and documentation will be uploaded and committed to the GitHub repository. Both team members will have a local copy of project deliverables to be turned in at the end of the course.

4. Technical Process Plans

4.1. Process Model

The Escape Sequence project follows an iterative and incremental development model. This approach allows the team to develop the game in stages, gradually adding features and improving the system through repeated cycles of design, implementation, and testing. Because the project is completed within an academic year and by a small development team, an iterative approach allows the team to identify problems early and make improvements during development.

The development process is divided into several major phases. Each phase produces deliverables that are reviewed before the project moves forward.

Iteration 1 – Planning and Requirements

During this phase, the team defines the project scope, develops the project proposal, and creates the Software Requirements Specification (SRS). Game mechanics, system requirements, and user interface expectations are established during this stage.

Iteration 2 – System Design

The second phase focuses on system architecture and design. The team creates diagrams such as use case diagrams and activity diagrams and defines the internal structure of the system, including classes for players, cards, decks, and game logic.

Iteration 3 – Implementation

During this phase, the development team implements the core functionality of the game. This includes programming the deck structure, player interactions, specialty card behavior, and the graphical user interface.

Iteration 4 – Testing and Debugging

The system is tested to verify that game mechanics, user interface actions, and gameplay rules function correctly. Bugs and logic errors are identified and corrected during this stage.

Iteration 5 – Final Integration and Delivery

In the final phase, the completed system is finalized, documentation is reviewed, and the project is prepared for demonstration and submission.

Management review checkpoints occur at the following milestones:

- Project proposal submission
- Software Requirements Specification (SRS) submission
- Design review milestone
- Implementation progress review
- Final project presentation and demonstration

4.2. Methods, Tools, and Techniques

The Escape Sequence project uses a combination of software development methodologies, programming tools, and testing practices to design, implement, and validate the system.

The system is developed using the following hardware and operating environments:

- **Hardware:** Standard personal computers or laptops capable of running Java development tools
- **Operating Systems:** Windows, macOS, or Linux
- **Runtime Environment:** Java Runtime Environment (JRE)

The following software tools support development activities:

- **NetBeans IDE:** Primary development environment for coding, debugging, and compiling Java programs.

- **Java Development Kit (JDK):** Provides libraries and tools needed to build and run Java application.
- **Java Swing:** Used for building the graphical user interface
- **Text and document editors:** Used to prepare documentation and project reports
- **Diagram tools:** Used to create system diagrams such as use case and activity diagrams

The project follows an iterative software development methodology in which the system is developed in stages. Requirements are defined first, followed by design, implementation, and testing activities.

Key development practices include:

- Object-oriented design
- Modular programming using classes and methods
- Clear separation of system components such as game logic, interface, and data storage
- Consistent documentation of code and system design

Several quality assurance techniques are used during development:

- **Code reviews** between team members to identify errors and improve code quality
- **Unit testing** to verify that individual components of the game function correctly
- **Integration testing** to ensure different system components interact properly
- **Debugging tools within NetBeans IDE** to detect and correct programming errors
- **System testing** to verify that the complete game behaves according to the requirements

4.3. Configuration Management Plan

Configuration management is used in the Escape Sequence project to ensure that project artifacts such as source code, documentation, and diagrams are properly organized and controlled throughout development.

Because the project team is small, configuration management activities are handled internally by the development team.

The following project artifacts are managed under configuration control:

- Source code files
- Software Requirements Specification (SRS)
- Project Management Plan
- System design diagrams
- Testing documentation
- Final project deliverables

Changes to the system or documentation are discussed and approved by both team members before implementation. This ensures that updates are understood and agreed upon by the development team.

Significant changes to system functionality or project requirements are documented and incorporated into updated project documents when necessary.

Version tracking is maintained by labeling document revisions and storing updated files in the project repository or shared project storage. This allows the team to monitor which version of a document or code file is the most current.

Periodic reviews are conducted to ensure that documentation, code, and project artifacts remain consistent with project requirements. These reviews occur during milestone checkpoints such as SRS completion, design review, and final submission.

The final version of the Escape Sequence software will be packaged with all required documentation and submitted as the final project deliverable.

Configuration management activities are expected to represent approximately 5–10% of the total project effort, including version tracking, document updates, and artifact organization.

Responsibility for configuration management is shared between the team members, with Akera Griffith coordinating documentation version updates and Kaitlyn Morris assisting with software version control and testing documentation updates.

4.4. Quality Assurance Plan

The purpose of the Quality Assurance (QA) plan for the Escape Sequence project is to ensure that the software system meets its functional and nonfunctional requirements and performs reliably for users. Quality assurance activities are integrated throughout the development lifecycle to identify defects early, verify system behavior, and ensure that documentation and implementation remain consistent with project requirements.

The QA process includes several activities such as peer reviews, testing, verification and validation, and configuration management. Peer reviews are conducted between team members to examine code, documentation, and design decisions before implementation or submission. This helps detect logical errors, inconsistencies, or missing requirements early in development.

Testing activities include unit testing, integration testing, and system testing. Unit testing verifies that individual program components such as card logic, player actions, and specialty card behavior function correctly. Integration testing ensures that different system components, such as the graphical user interface and game logic, interact properly. System testing evaluates the complete game to confirm that it operates according to the requirements defined in the Software Requirements Specification (SRS).

Configuration management activities support quality assurance by maintaining consistent versions of documentation and source code. Version control and document updates help ensure that team members are working with the most current project artifacts.

The primary QA responsibilities are divided between the two team members. Kaitlyn Morris primarily manages testing and debugging activities, while Akera Griffith supports peer review and documentation quality checks. Both members collaborate on verifying system functionality and resolving defects.

Quality assurance activities are estimated to represent approximately 15–20% of the total project effort, including code review, testing, debugging, and documentation validation. These activities are incorporated into the project schedule and occur throughout the development process, particularly during the implementation and testing phases.

4.5. Documentation Plan

The Escape Sequence project requires several types of documentation to support development, testing, and final delivery of the software system. These documents provide detailed information about system requirements, design decisions, installation procedures, and gameplay instructions for users.

The primary documentation deliverables include the project proposal, Software Requirements Specification (SRS), Project Management Plan, user guide, installation guide, testing report, and final

project documentation. These documents help ensure that the system can be understood, used, and maintained effectively.

All documentation will be stored digitally with the project files and shared between team members during development. Final versions will be submitted through the course submission platform as required by the instructor.

Document	Template or Standard	Created By	Reviewed By	Target Date	Distribution
Project Proposal	Course proposal format	Akera Griffith, Kaitlyn Morris	Dr. Zhao	Sept. 9, 2025	Course submission
Software Requirements Specification (SRS)	IEEE 830 SRS format	Akera Griffith, Kaitlyn Morris	Dr. Zhao	Oct. 20, 2025	Course submission
Project Management Plan	Course PMP template	Akera Griffith, Kaitlyn Morris	Dr. Zhao	Mid-semester milestone	Course submission
System Design Documentation	UML diagram standards	Akera Griffith, Kaitlyn Morris	Dr. Zhao	During development	Project files
User Guide	Game instruction manual format	Kaitlyn Morris	Akera Griffith	Before final submission	Included with final project
Installation Guide	Basic installation instructions	Akera Griffith	Kaitlyn Morris	Before final submission	Included with project
Testing and Debugging Report	Course testing documentation format	Kaitlyn Morris	Akera Griffith	End of semester	Course submission

Release Notes	Version summary format	Akera Griffith	Kaitlyn Morris	Final submission	Included with project
Final Project Documentation	Course report format	Akera Griffith, Kaitlyn Morris	Dr. Zhao	End of semester	Course submission

4.6. Process Improvement Plan

The Escape Sequence project includes small process improvements to improve development efficiency and organization. One improvement involves refining documentation templates for project artifacts such as the SRS and project reports to maintain consistent formatting and clarity. Another improvement focuses on structured task tracking, where the team regularly reviews progress and assigns tasks during meetings to ensure milestones are met. The team will also improve testing practices by identifying bugs earlier through regular code review and testing during development.

Throughout development, the team will monitor how these process improvements affect project progress. If any approach proves inefficient or creates unnecessary delays, it will be modified to better support the project workflow. The results of these improvements will be reviewed during the project retrospective, where the team will capture lessons learned regarding documentation practices, task organization, and testing strategies. These insights will help guide improvements in future software development projects.

Revision History

Name	Date	Reason for Changes	Version
Kaitlyn Morris	2/18/26	Revising Template	1.0 draft 1
Akera Griffith	3/6/26	Documentation	1.0 draft 1

